MAT 3007 – Optimization

The Simplex Method

Lecture 07

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Logistics

- Assigment 1 is due tomorrow, February 18th, 11:00 p.m.
- Assignment 2 is due on Monday, February 24th, 11:00 p.m.
- Midterm is scheduled on Saturday, March 22nd, 10:00 a.m. – 11:30 a.m.

Recap

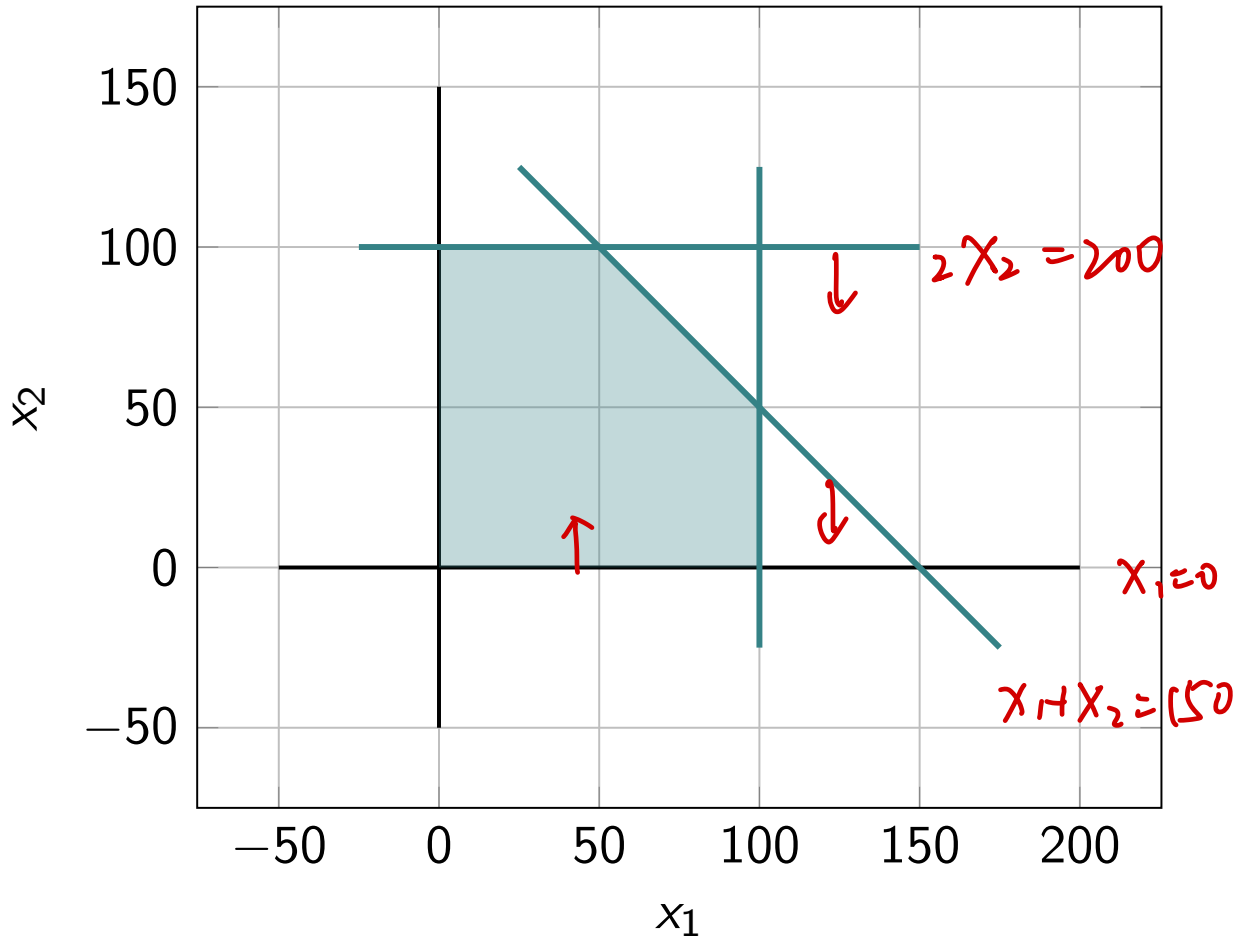
Starting Point: Graphical Solutions to LP

Recall the production problem:

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0.
\end{array}
$$

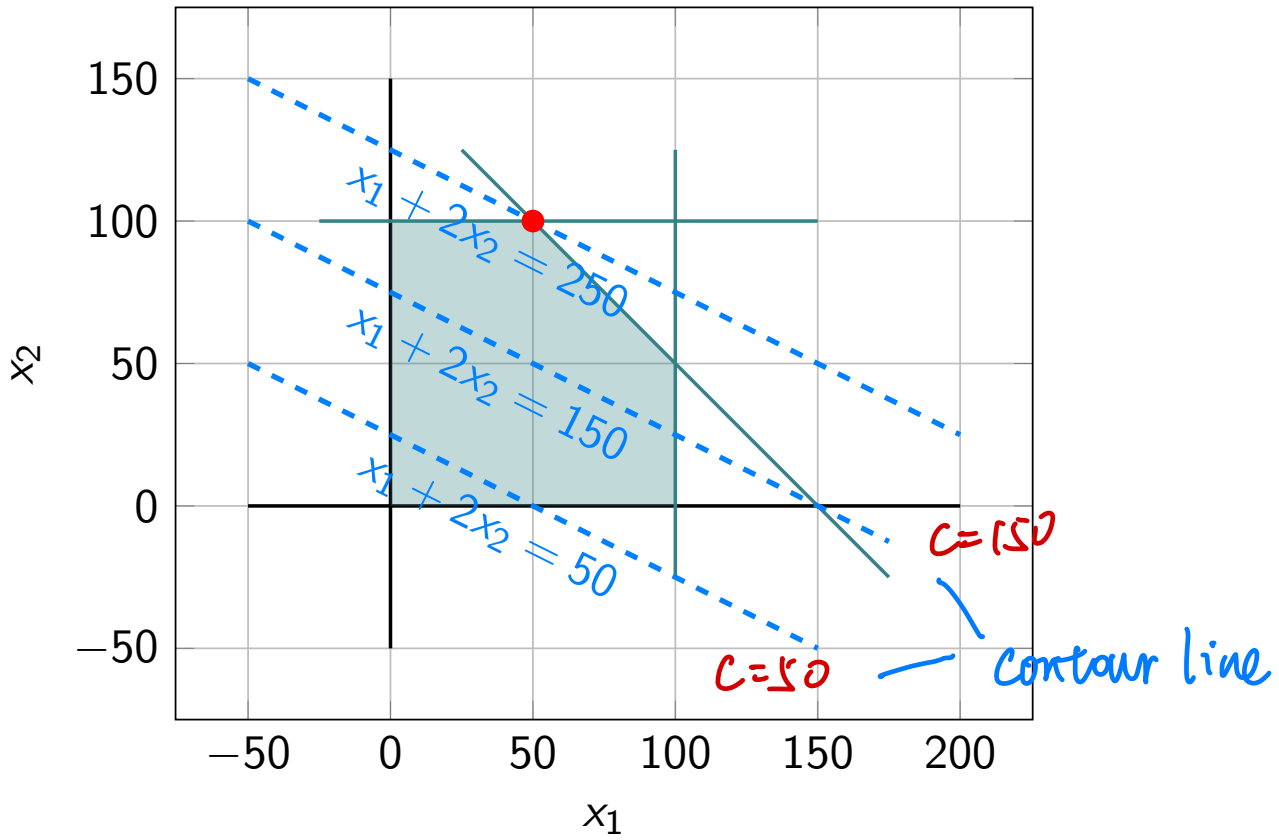
Solve LP from Graph

We first draw the feasible region.



To Maximize $x_1 + 2x_2$ ...

We then draw the function $x_1 + 2x_2 = c$ for different values of c.



- The optimal solution is the highest one among these lines that touches the feasible region.

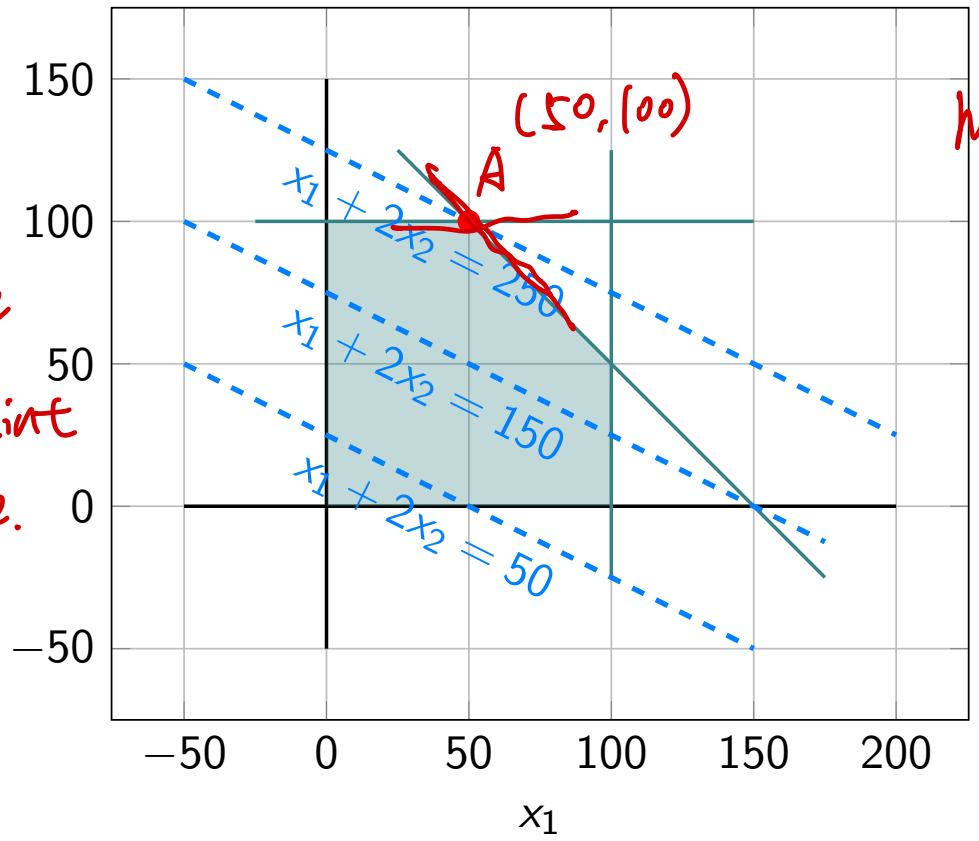
Some Observations

Observations:

- The feasible region of an LP is a polyhedron. $\{x : Ax \le b\}$
- The optimal solution tends to be a corner.
- At those "corners", some constraints are active, some are not.

active: the constraint hold with equality.

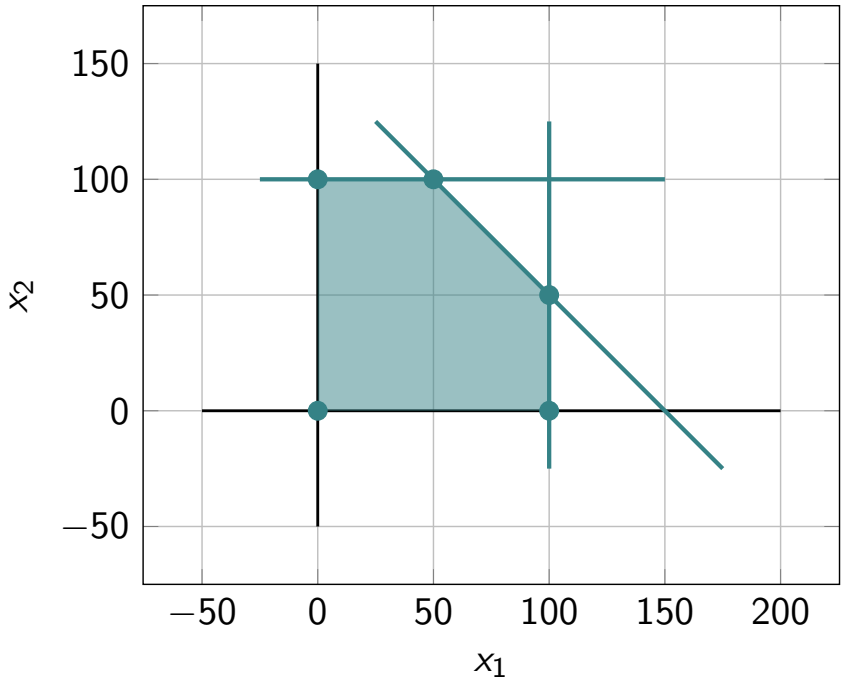
At point A, constraints $2x_2 \le 200$ and $x_1 + x_2 \le 150$ are active. But constraint $x_1 \le 100$ is not active.



Definition: Extreme Point

Let P be a polyhedron. A point $\boldsymbol{x} \in P$ is said to be an extreme point of P if we cannot find two vectors $\boldsymbol{y}, \boldsymbol{z} \in P$ with $\boldsymbol{y}, \boldsymbol{z} \neq \boldsymbol{x}$ and a scalar $\lambda \in [0, 1]$, such that $\boldsymbol{x} = \lambda \boldsymbol{y} + (1 - \lambda)\boldsymbol{z}$.

- That is, $\boldsymbol{x}$ cannot be represented as a convex combination of other two points in P.



Finding Extreme Points of an LP

In the following, we consider an LP in its standard form:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{A}$ is an $m \times n$ matrix ($m < n$) and $\boldsymbol{b} \in \mathbb{R}^m$.

General Assumption:

$\boldsymbol{A}$ has linearly independent rows.

the number of linearly indepent rows = the number of linearly independent columns.

⇒ we can find m linearly independent columns of A.

- Recall that $\text{rank}(\boldsymbol{A}) = \text{column-rank}(\boldsymbol{A}) = \text{row-rank}(\boldsymbol{A})$. Under this general assumption, $\text{rank}(\boldsymbol{A}) = m$ and there are m linearly independent columns.

Extreme Points of LPs: Basic Solutions

Definition: Basic Solution

We call $\boldsymbol{x}$ a basic solution of the LP if and only if

1. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
2. There exist indices $B(1), \ldots, B(m)$ such that the columns of

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

are linearly independent and $x_i = 0$ for $i \neq B(1), \ldots, B(m)$.

Finding a Basic Solution

Procedure to Find a Basic Solution:

1. Choose any m independent columns of $\boldsymbol{A}$: $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$.
2. Let $x_i = 0$ for all $i \neq B(1), \ldots, B(m)$. $\Leftrightarrow x_N = 0$
3. Solve the equation $\boldsymbol{Ax} = \boldsymbol{b}$ for the remaining $x_{B(1)}, \ldots, x_{B(m)}$. This is equivalent to $\boldsymbol{A}_B \boldsymbol{x}_B = \boldsymbol{b}$ or $\boldsymbol{x}_B = \boldsymbol{A}_B^{-1} \boldsymbol{b}$.

Define $N = \{1, 2, \ldots, n\} \setminus B$.

$$Ax = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n = \sum_{i \in B} x_i A_i + \sum_{j \in N} x_j A_j = [A_B \; A_N] \begin{pmatrix} x_B \\ x_N \end{pmatrix}$$

$$= A_B x_B + A_N x_N = A_B x_B$$

$x_N = 0$ because $x_i = 0$ for $i \notin B$.

Also note that matrix A_B is of dimension $m \times m$, and $\operatorname{rank}(A_B) = m \Rightarrow A_B$ invertible

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- Since $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$ are linearly independent, $\boldsymbol{A}_B$ is invertible and the last step has a unique solution.

Basic solution: after we choose B s.t $A_{B(1)} \ldots A_{B(m)}$ Lin. independent

$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \begin{pmatrix} A_B^{-1} b \\ 0 \end{pmatrix}$ is a BS

Basic Feasible Solutions

Definition: Basic Feasible Solution

If a basic solution $\boldsymbol{x}$ also satisfies that $\boldsymbol{x} \geq \mathbf{0}$, then we call it a basic feasible solution (BFS).

How to Find a BFS?

- First find a basic solution $\boldsymbol{x}$. $x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \begin{pmatrix} A_B^{-1} b \\ 0 \end{pmatrix}$
- Check if $\boldsymbol{x} \geq \mathbf{0}$. $\Longleftrightarrow$ $x_B \geq 0$

Theorem: Extreme Points and BFS

For the standard LP polyhedron $P := \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \mathbf{0}\}$, the following statements are equivalent:

1. $\boldsymbol{x}$ is an extreme point of P.
2. $\boldsymbol{x}$ is a basic feasible solution.

Example: Production Planning

Recall the production problem:

$$
\begin{aligned}
\text{maximize} \quad & x_1 + 2x_2 \\
\text{subject to} \quad & x_1 \leq 100 \\
& 2x_2 \leq 200 \\
& x_1 + x_2 \leq 150 \\
& x_1, \; x_2 \geq 0
\end{aligned}
$$

with standard form:

$$
\begin{aligned}
\text{minimize} \quad & -x_1 - 2x_2 \\
\text{subject to} \quad & x_1 + s_1 = 100 \\
& 2x_2 + s_2 = 200 \\
& x_1 + x_2 + s_3 = 150 \\
& x_1, \; x_2, \; s_1, \; s_2, \; s_3 \geq 0
\end{aligned}
$$

Example: Continued

We can denote the feasible set by $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix}.$$

- Choose 3 independent columns of $\boldsymbol{A}$, e.g., the first 3 ($B = \{1, 2, 3\}$), we get the corresponding basic solution via:

$$\boldsymbol{x}_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix}.$$

- That is $x_1 = 50, x_2 = 100, s_1 = 50$. Hence, $(50, 100, 50, 0, 0)$ is a basic solution. One can find other basic solutions by choosing other sets of columns.

Example Continued

choose 3 columns out of 5 columns

$C(5,3) = \frac{5\times4\times3}{3\times2} = 10$

We can list all basic solutions:

Indices	$\{1,2,3\}$ (A)	$\{1,2,4\}$ (B)	$\{1,2,5\}$ (C)
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Status	feasible	feasible	not feasible

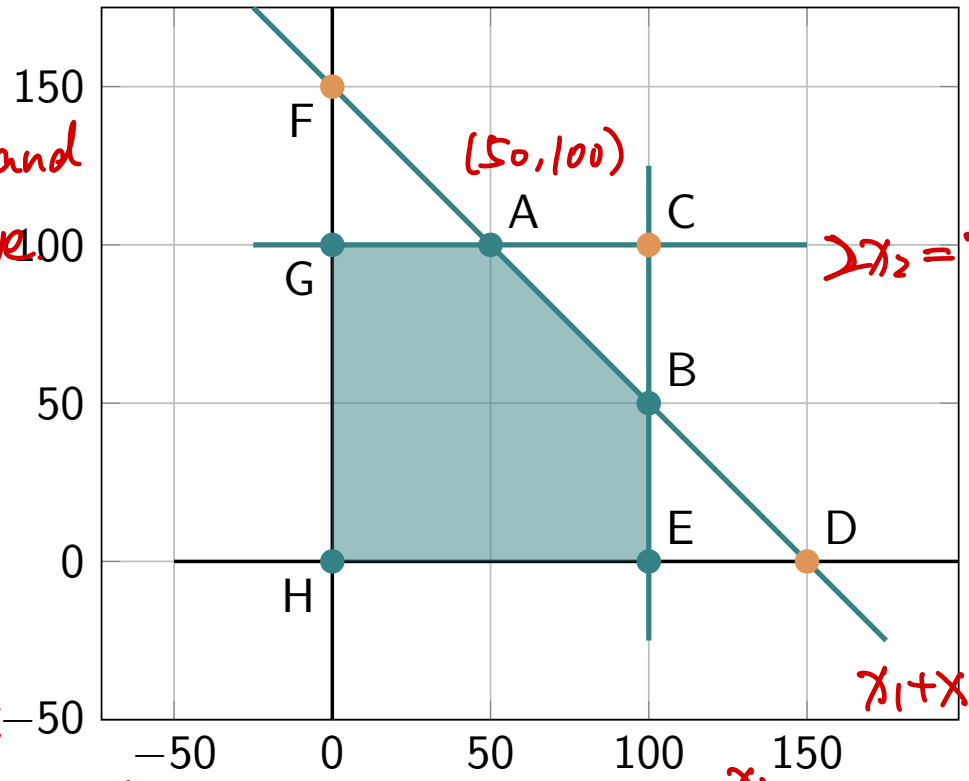
Indices	$\{1,3,4\}$ (D)	$\{1,4,5\}$ (E)	$\{2,3,4\}$ (F)
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Status	not feasible	feasible	not feasible

Indices	$\{2,3,5\}$ (G)	$\{3,4,5\}$ (H)	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Status	feasible	feasible	

The other two choices $\{1,3,5\}$ and $\{2,4,5\}$ lead to dependent basic columns (therefore no basic solutions can be obtained).

8 BSs : 5 BFSs

Example Continued



At point A.

constraint $2x_2 \leq 200$ and $x_1 + x_2 \leq 150$ are active

$\Rightarrow$ slack variables $s_2 = 0$, $s_3 = 0$

Also, $x_1 > 0$, $x_2 > 0$.

And the other constraint $x_1 \leq 100$ is not active

$\Rightarrow s_1 > 0$.

Therefore, we conclude $\begin{pmatrix} x_1 \\ x_2 \\ s_1 \end{pmatrix}$ are basic variables at A by simply observing the graph.

- The BSs (in $\mathbb{R}^2$) correspond to all possible intersections of two constraints.
- The BFSs A, B, E, G, H correspond to all extreme points.

Fundamental LP Theorem

Consider a linear problem in standard form and assume that $\boldsymbol{A}$ has full rank m.

1. If the feasible set is nonempty, there is a BFS.
2. If there is an optimal solution, there is an optimal solution that is also a BFS.

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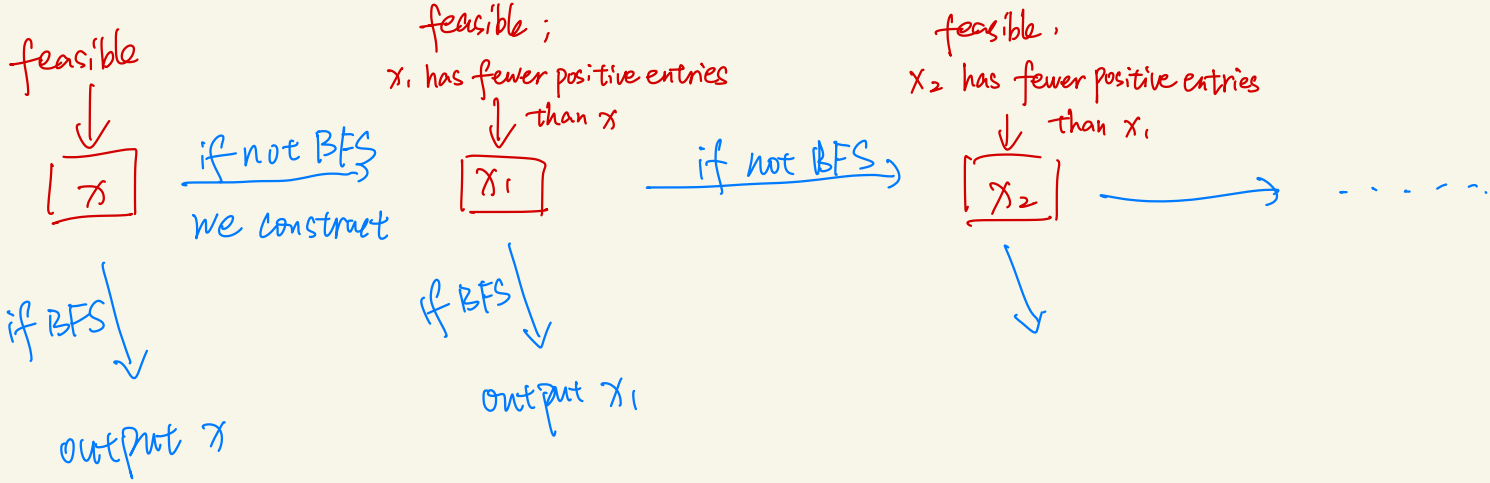
$\rightsquigarrow$ In order to find an optimal solution, we only need to look among BFSs!

Corollary: Characteristics

If an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution with no more than m positive entries.

all BSs have no more than m positive entries

proof idea of part (i). Starting from a feasible solution x.



consider this as an algorithm: ① output is a BFS

② this algorithm stops in finite steps.

Proof Sketch

We first prove part (i). Assume there is a feasible solution $\mathbf{x}$. Let B denote the index set for which $\mathbf{x}$ is positive, and let k be the size of B. In the following, we will show that if $\mathbf{x}$ is not a basic solution, we can always find a solution that has an index set with size less than k. By repeatedly doing this we must be able to find a basic feasible solution.

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why?).

If they are linearly independent, $k \le m$ because $\text{rank}(A) = m$.

— Case 1 ($k=m$). We have m linearly indep. colmns, and $x_i = 0$ for $i \notin B$.

$\Rightarrow$ x is a BS $\rightarrow$ Contradiction

— Case 2 ($k<m$). Because $\text{rank}(A) = m$, we can find a set $S = \{S(1), \ldots, S(m-k)\}$ such that $A_{B(1)}, \ldots, A_{B(m)}, A_{S(1)}, \ldots A_{S(m-k)}$ are linearly indep. Also, $x_i = 0$ for $i \notin \{B \cup S\}$. $\Rightarrow x$ is a BS

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why?). There exists $\{\{\beta_i\}_{i\in B}\}$ such that $\sum_{i\in B} \beta_i A_i = 0$ and at least one β_i is positive (why?).

if all $\beta_i \le 0$ for $\forall i$, then $\sum_{i\in B} (-\beta_i) A_i = 0$, so we can just use $-\beta$ as coefficient.

Proof Sketch

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To show that, if $\mathbf{x}$ is not a basic solution, then it must be that $\{A_i, i \in B\}$ are linearly dependent (why?). There exists $\{\{\beta_i\}_{i\in B}\}$ such that $\sum_{i\in B} \beta_i A_i = 0$ and at least one β_i is positive (why?). Define a vector $\boldsymbol{\alpha}$ by

$$\alpha_i = \begin{cases} \beta_i & i \in B \\ 0 & \text{otherwise} \end{cases}.$$

Then we consider $\mathbf{x} - \epsilon\boldsymbol{\alpha}$. There must exist an ϵ such that it is still feasible and that there is at least one more 0 entry in $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ (why?). Thus part (i) is proved.

Let choose $\epsilon = \min_i \left\{ \frac{x_i}{\alpha_i} : i \in B, \alpha_i > 0 \right\}$ (well-defined because there exists i such that $\alpha_i = \beta_i > 0$)

define $N = \{1, \dots, n\} \setminus B$

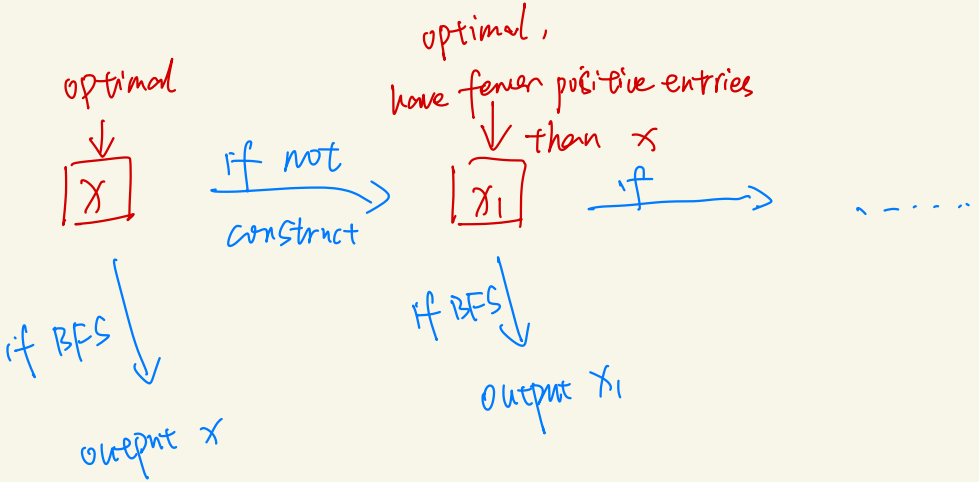
— $A(x - \epsilon\alpha) = \underbrace{Ax}_{=b} - \epsilon \underbrace{A\alpha}_{} = b$

$= [A_B \; A_N] \begin{pmatrix} \alpha_B \\ \alpha_N \end{pmatrix} = A_B\alpha_B + \underbrace{A_N\alpha_N}_{=0} = A_B\alpha_B = A_B\beta = 0$

— Then look at

$$x_i - \epsilon\alpha_i = \begin{cases} x_i = 0 & \text{if } i \in N \\ 0 & \text{if } \epsilon = \frac{x_i}{\alpha_i} \\ > 0 & \text{for other } i \end{cases}$$

Proof idea for part (ii),



we construct $x - \epsilon\alpha$ exactly the same way as part (i). Now we want to state $c^T\alpha = 0$. (why?)

observe that $x + \tilde{\epsilon}\alpha$ and $x - \tilde{\epsilon}\alpha$ are feasible for $\tilde{\epsilon} > 0$ small enough

Then — if $c^T\alpha > 0$, then $c^T(x - \tilde{\epsilon}\alpha) = c^Tx - \tilde{\epsilon}c^T\alpha < c^Tx$ → contradiction

— if $c^T\alpha < 0$, then $c^T(x + \tilde{\epsilon}\alpha) = c^Tx + \tilde{\epsilon}c^T\alpha < c^Tx$ → contradiction.

Since $c^T d = 0$, the new constructed point $x - \epsilon d$ has objective.

$$c^T(x - \epsilon d) = c^T x - \epsilon \underbrace{c^T d}_{=0} = c^T x.$$

$\Rightarrow$ $x - \epsilon d$ is also optimal.

The rest of the proof idea is the same as part (i).

Proof Sketch Continued

Now we consider part (ii). We continue to use the arguments for part (i). Let $\boldsymbol{x}$ be an optimal solution with k positive entries. If x is not a basic solution, then we show that we can always find an optimal solution with fewer than k positive entries (or reach a contradiction). By repeatedly doing this, we must be able to find an optimal basic feasible solution.

We continue to define the same $\boldsymbol{\alpha}$. We claim that it must be that $\mathbf{c}^T\boldsymbol{\alpha} = 0$, otherwise, one can construct a better solution than $\mathbf{x}$ by considering $\mathbf{x} + \epsilon\boldsymbol{\alpha}$ and $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ for some small ϵ, which contradicts with the optimality of $\mathbf{x}$.

If $\mathbf{c}^T\boldsymbol{\alpha} = 0$, then $\mathbf{x} - \epsilon\boldsymbol{\alpha}$ are still optimal solutions. And by previous arguments, we can find one optimal solution with fewer positive entries. $\square$

From Basic Feasible Solutions to Optimal Solutions



Consequences:

- We only need to search among basic feasible solutions to find the optimal solution.

From Basic Feasible Solutions to Optimal Solutions

Consequences:

- We only need to search among basic feasible solutions to find the optimal solution.

How can we search among the basic feasible solutions?

- One may suggest to list all the BFSs and compare their objective values.

$\rightsquigarrow$ There might be too many BFS!

- For a linear optimization with m constraints and n variables, how many basic feasible solutions can we have?

$\rightsquigarrow$ $C(n, m)$ ... if $n = 1000$, $m = 100$, then this number is 10^{143} ...

The Simplex Method

The simplex method proceeds from one BFS (a corner point of the feasible region) to a neighboring one to continuously improve the value of the objective function until reaching optimality.

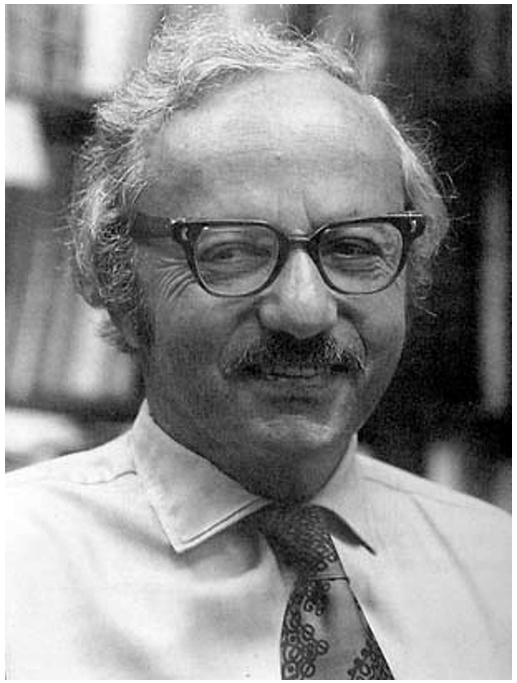


Figure 1: The simplex method was invented by George Dantzig.

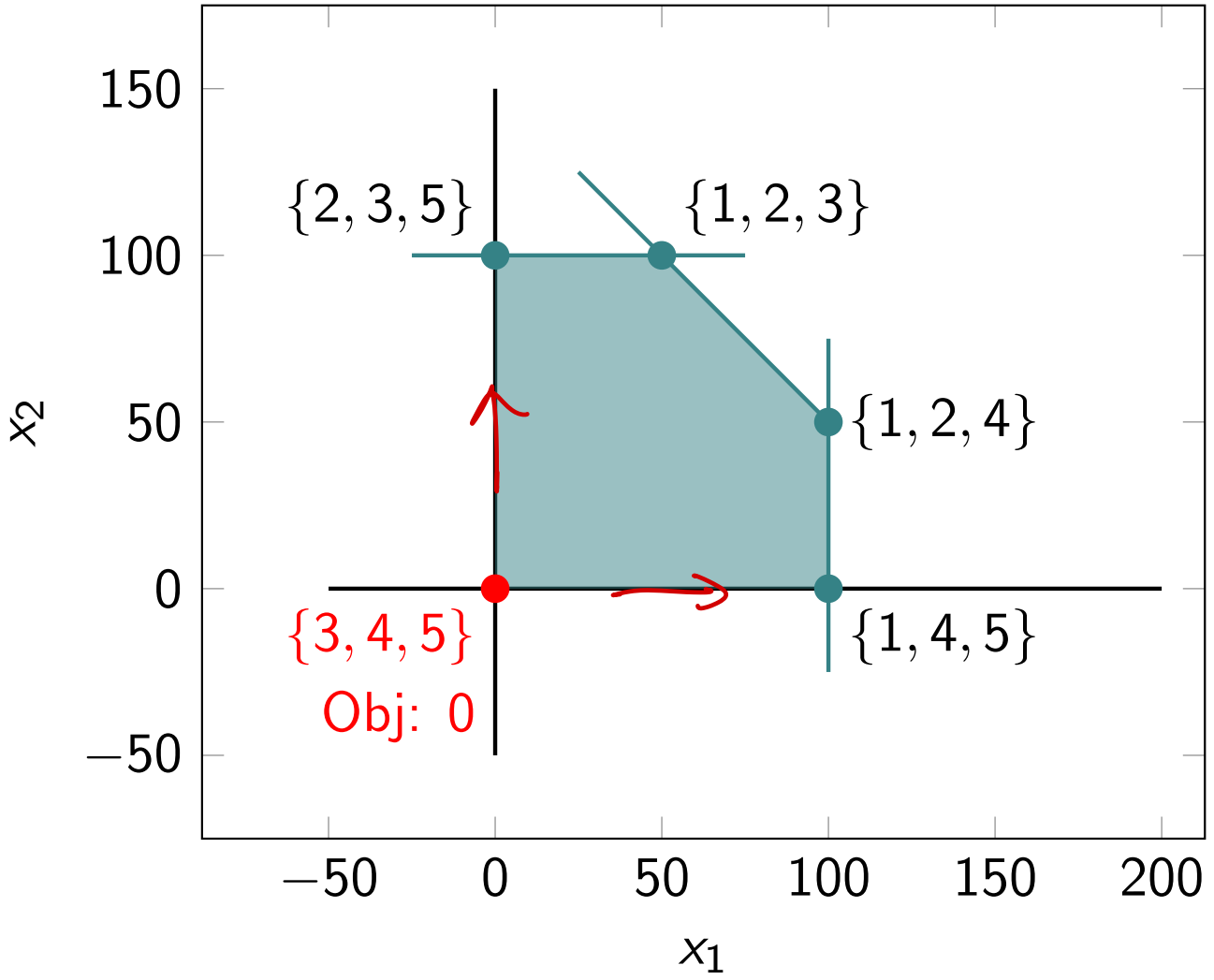
Neighboring BFS

Definition: Neighboring Basic Solutions

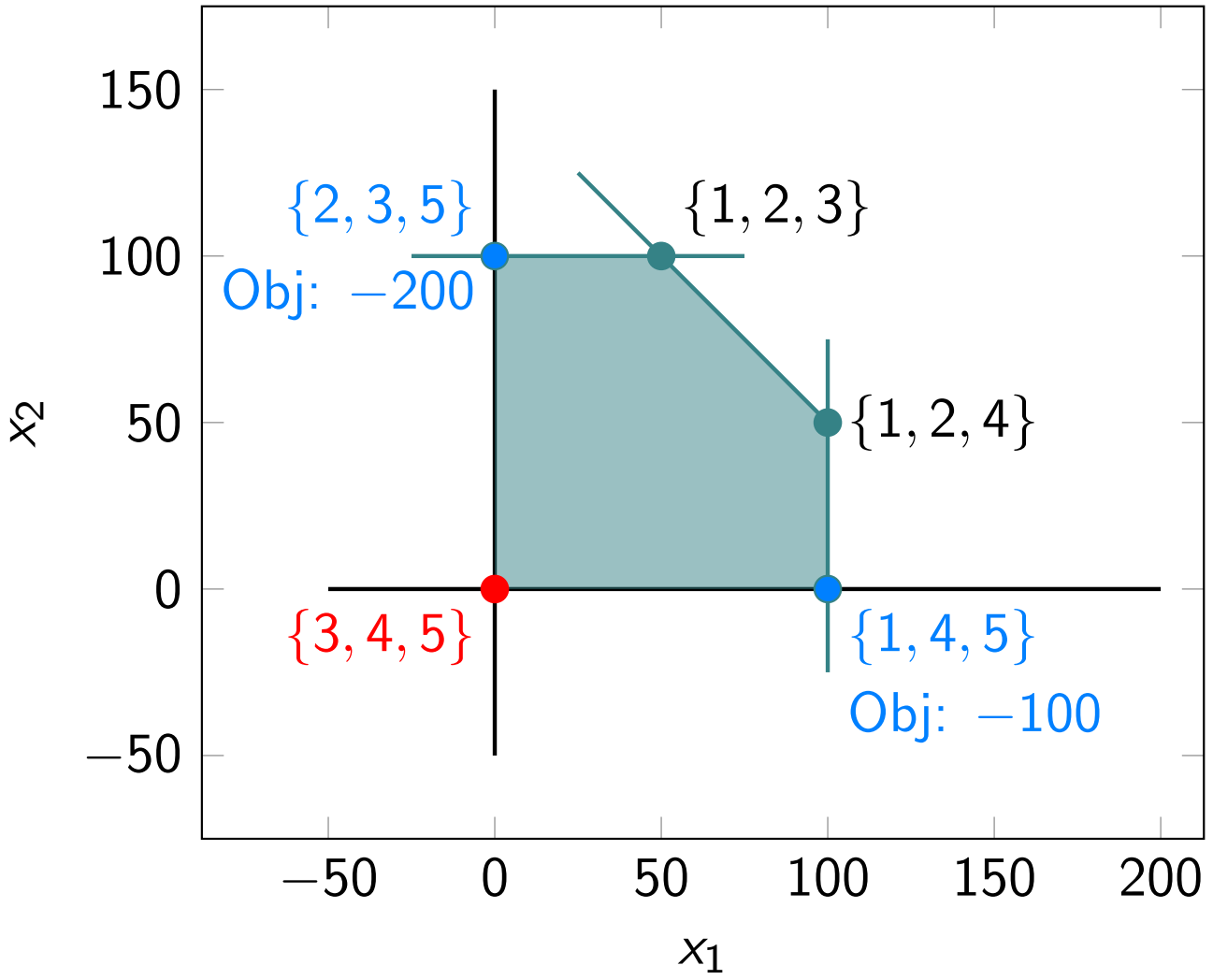
Two basic solutions are neighboring (or adjacent) if they differ by exactly one basic (or non-basic) index.

- For example, a BFS constructed by using the columns $\{1, 2, 3\}$ is a neighbor to the BFS constructed by using the columns $\{1, 3, 5\}$ (but not $\{1, 4, 5\}$).

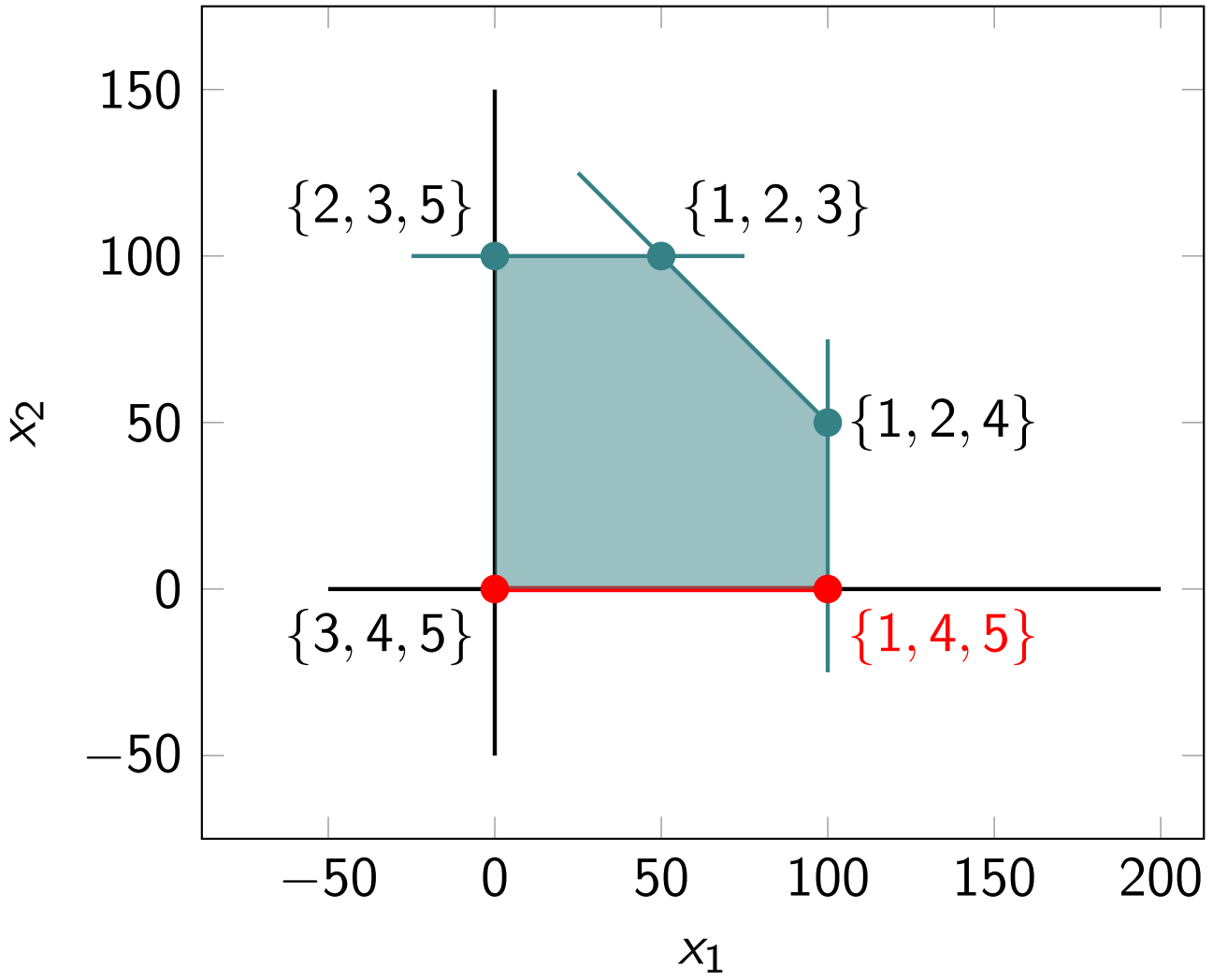
Example



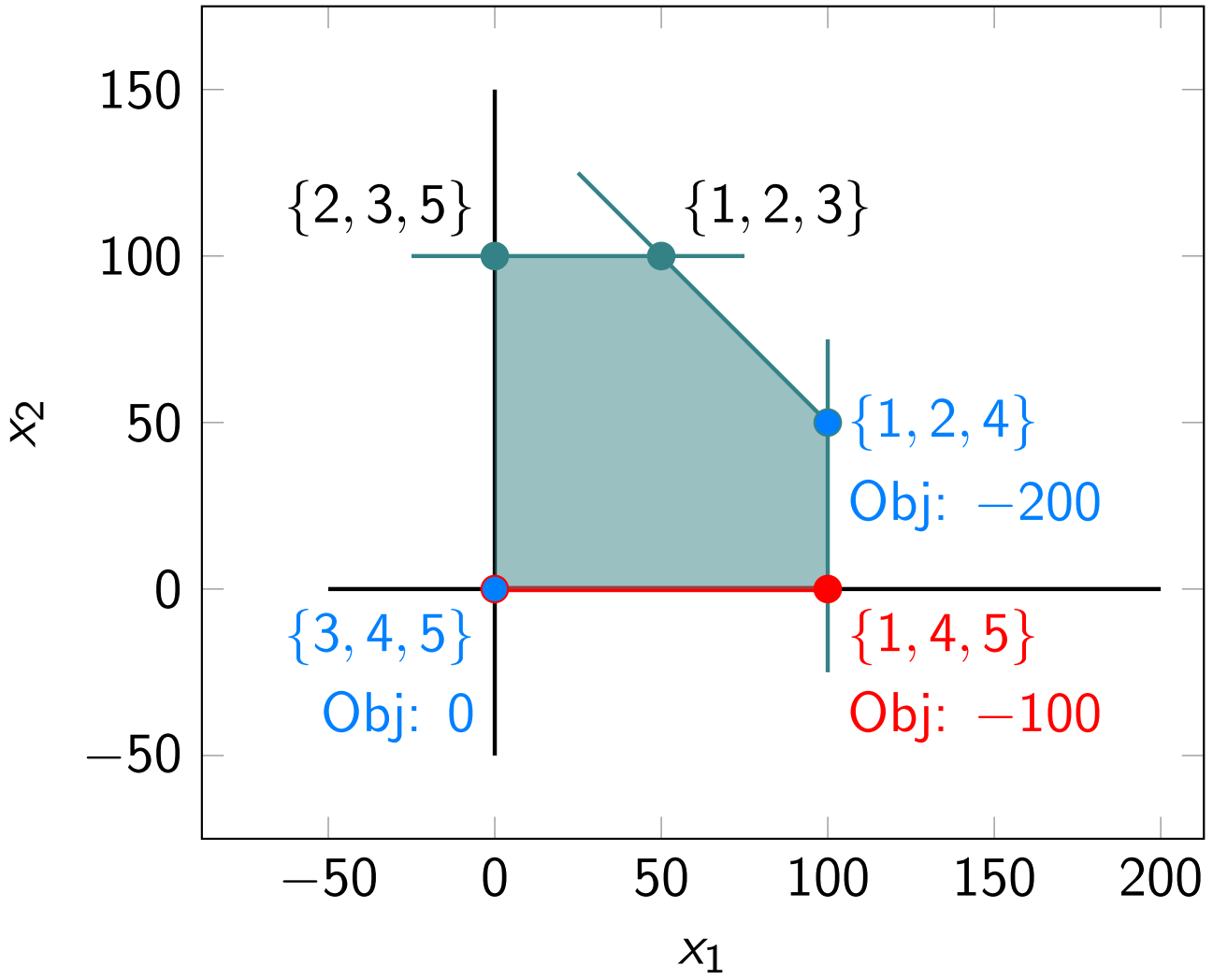
Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
Obj: −200
{1, 2, 3}
{1, 2, 4}
{3, 4, 5}
{1, 4, 5}
Obj: −100

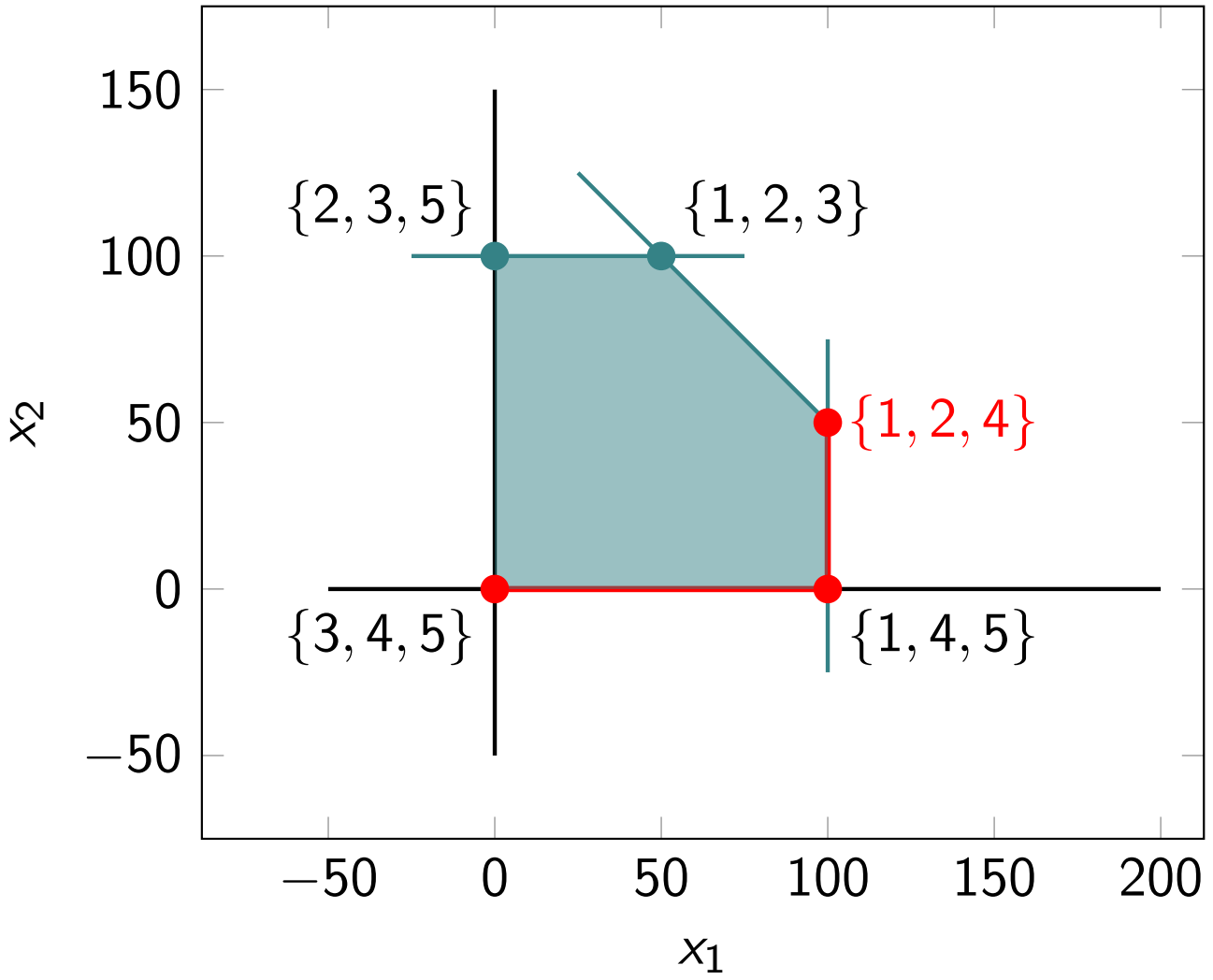
Example



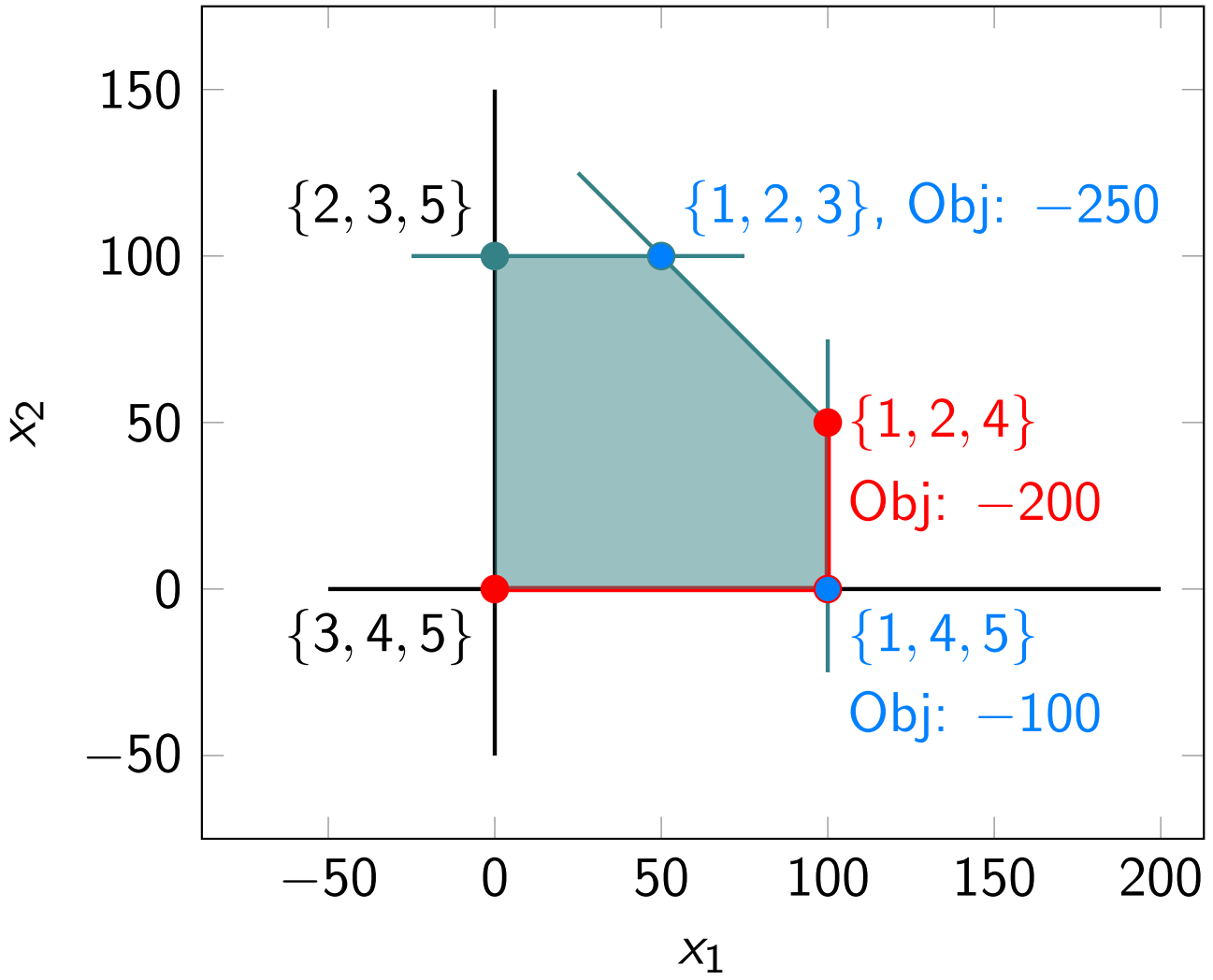
Example


150
100
50
0
−50
−50
0
50
100
150
200
x_2
x_1
$\{2,3,5\}$
$\{1,2,3\}$
$\{1,2,4\}$
Obj: -200
$\{3,4,5\}$
Obj: 0
$\{1,4,5\}$
Obj: -100

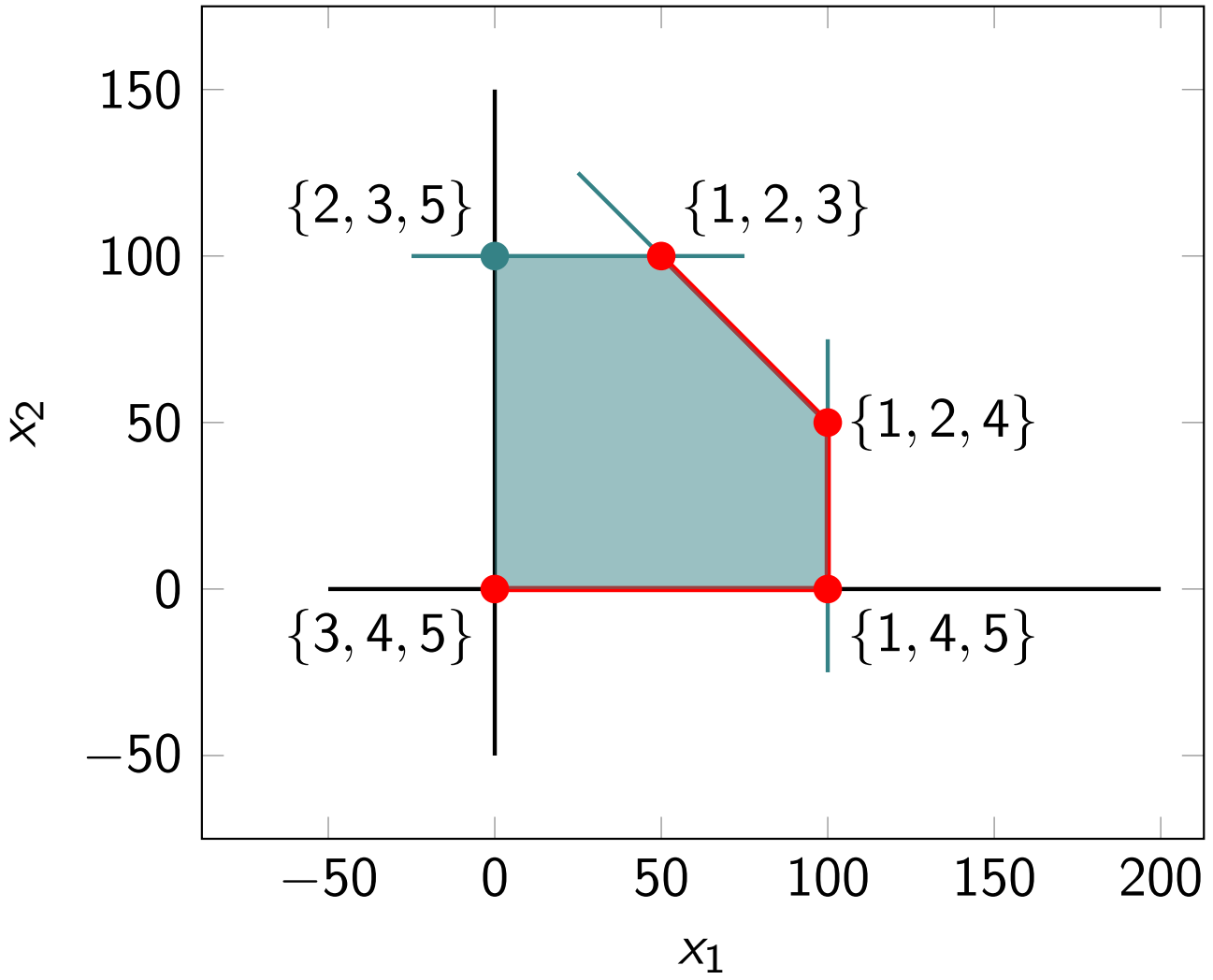
Example



Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
{1, 2, 3}, Obj: −250
{1, 2, 4}
Obj: −200
{3, 4, 5}
{1, 4, 5}
Obj: −100

Example



Checking Neighbors

The main question is how to find a neighboring BFS that improves the current one (reduces the objective function):

A naïve way is simply to check all neighbors of the current BFS:

- There are $m(n-m)$ potential neighbors for a BFS (choosing one index to leave, choosing one to enter). For each one, we need to solve a linear equation of m variables.

want one non-basic variable to be basic — (n-m) choices

want one basic variable to be non-basic — m choices

Finding and Moving to Neighboring BFS

Simplex Method: Starting Point

Starting from a BFS $\boldsymbol{x}$ with basis is $B(1), \ldots, B(m)$,

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

Denote N the non-basic index set. B and N are disjoint, $|B| = m$, $|N| = n - m$, and $B \cup N = \{1, 2, ..., n\}$. Let $\boldsymbol{A}_N$ be the matrix consisting of the non-basic columns of $\boldsymbol{A}$.

Rearranging the variables, we can write $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, $\boldsymbol{x} = [\boldsymbol{x}_B; \boldsymbol{x}_N]$, where $\boldsymbol{x}_B$ are the basic variables, and $\boldsymbol{x}_N$ are the non-basic variables.

$$\boldsymbol{x}_B = \boldsymbol{A}_B^{-1}\boldsymbol{b} \geq \boldsymbol{0}, \qquad \boldsymbol{x}_N = 0.$$

Finding a Neighboring BFS

We want to find a neighboring BFS:

- Finding a neighbor means changing one of the basic indices.
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Finding a Neighboring BFS

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We want a non-basic variable x_j (with $j \in N$) to be basic. We consider moving $\boldsymbol{x}$ (the current BFS) to $\boldsymbol{x} + \theta \boldsymbol{d}$, $\theta \geq 0$, where:

1. $d_j = 1$. ← increase x_j from 0 to positive
2. $d_{j'} = 0$ for all other non-basic indices. ← other nonbasic variables stay 0.
3. How to choose $d_{j'} = 0$ for basic indices, i.e., $\boldsymbol{d}_B$?

$d_N = e_j \triangleq \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ where 1 is at j-th coordinate

What constraints do we have on $\boldsymbol{d}$? First, We need to guarantee that the resulting new point $\boldsymbol{x} + \theta \boldsymbol{d}$ is feasible:

$$\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b} = \boldsymbol{A}\boldsymbol{x} \quad \Longrightarrow \quad \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}.$$

because x is feasible

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Since $d_j = 1$ and $d_{j'} = 0$ for all other non-basic indices,

$$\boldsymbol{0} = \underbrace{\begin{bmatrix} \boldsymbol{A}_B & \boldsymbol{A}_N \end{bmatrix} \begin{pmatrix} \boldsymbol{d}_B \\ \boldsymbol{d}_N \end{pmatrix}}_{\boldsymbol{A}\boldsymbol{d}} = \boldsymbol{A}_B \boldsymbol{d}_B + \boldsymbol{A}_j. \xRightarrow{\boldsymbol{A}_B \text{ is invertible}} \boldsymbol{d}_B = -\boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

$$A_N d_N = \underbrace{d_j A_j}_{= A_j} + \underbrace{\sum_{\substack{i \in N \\ i \neq j}} d_i A_i}_{=0} = A_j$$

Basic Directions

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$$\boldsymbol{0} = \begin{bmatrix} \boldsymbol{A}_B & \boldsymbol{A}_N \end{bmatrix} \begin{pmatrix} \boldsymbol{d}_B \\ \boldsymbol{d}_N \end{pmatrix} = \boldsymbol{A}_B \boldsymbol{d}_B + \boldsymbol{A}_j. \Longrightarrow \boldsymbol{d}_B = -\boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

This means that the direction $\boldsymbol{d}$ is uniquely determined:

$$\boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N] = [-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0],$$

$\boldsymbol{d}_B$ $\boldsymbol{d}_N$

where the 1 is at the j-th entry. We call such $\boldsymbol{d}$ the j-th basic direction.

Basic Directions

We also need to consider the constraint $\boldsymbol{x} \geq \mathbf{0}$. We need to verify that this still holds (i.e., $\boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}$):

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- For non-basic variables i, x_i was 0 and d_i is non-negative (either 1 or 0), therefore, $x_i + \theta d_i$ is still non-negative.
- For basic variables, as long as they were all strictly positive, there must exist a small θ such that $\boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}$.

$$x_i + \theta d_i = \begin{cases} 0 & \text{if } i \in N,\ i \neq j \\ \theta & \text{if } i = j \\ > 0 & \text{if } i \in B \text{ and } x_i > 0 \text{ and } \theta \text{ small enough} \end{cases}$$

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Observation:

- Typically, the basic variables of a BFS are all positive (i.e., m positive entries).
- There are cases that some basic variables in a BFS are equal to 0 (we call it "degeneracy"). We will discuss it later.